



Strategi Peningkatan Loyalitas Pelanggan E-Commerce Menggunakan Analisis RFM pada Perusahaan XYZ

Group 4

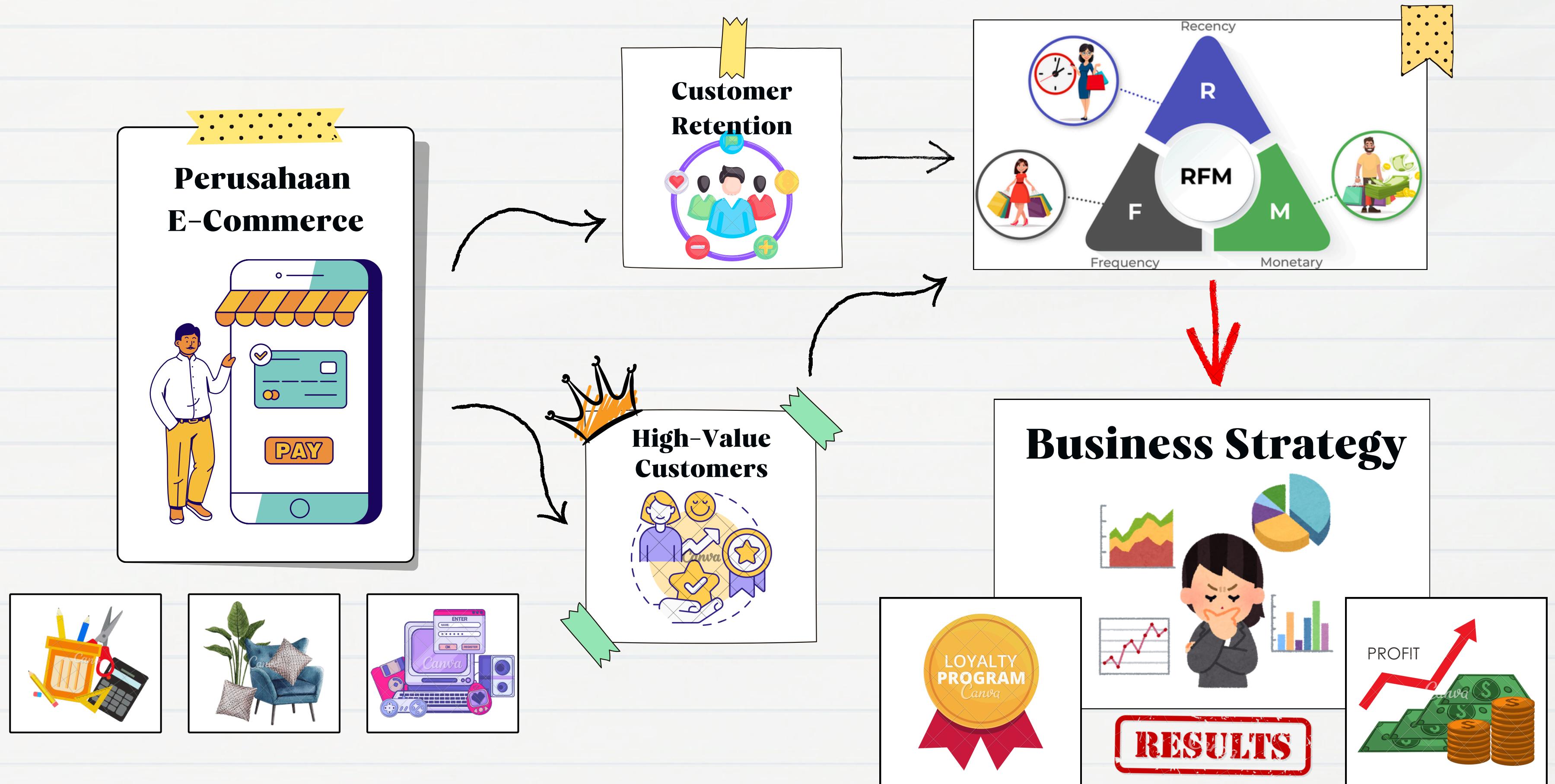
- Faaizah Faradhilah (007)-
 - Andi Kahfi (011)-
 - Rina Adibah (019)-
- 
- 

Overview

- Background
- Problem
- Tools & Materials
- Data Understanding
- Data Analysis
- Recommendation
- Conclusions



Background



Problems



1

Siapa saja **pelanggan terbaik** yang perlu mendapatkan perhatian khusus dari perusahaan?

2

Bagaimana cara **meningkatkan loyalitas pelanggan** yang berpotensi untuk menjadi lebih setia?

3

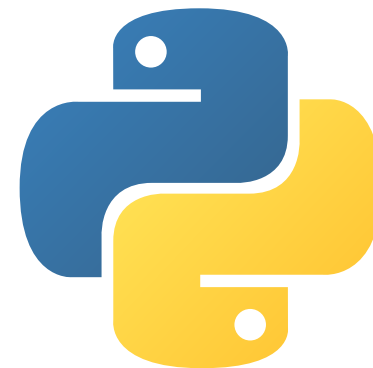
Pelanggan mana yang beresiko untuk tidak kembali berbelanja, dan bagaimana **cara memulihkan** mereka?

Tools & Materials

Dataset Sales Perusahaan XYZ
2011 - 2014

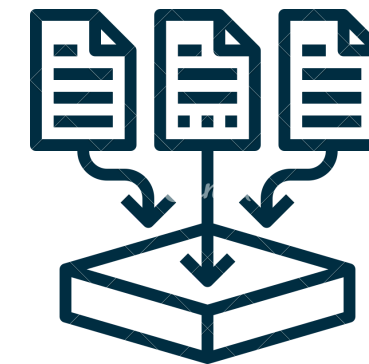


Libraries



```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

Kolom-kolom utama yang akan
digunakan



```
'Order Date': 'Recency',
'Order ID': 'Frequency',
'Sales': 'Monetary'
```



Data Collection

	Frequency	Recency	Customer Name	Monetary
0	BN-2011-7407039	2011-01-01	Ruby Patel	45
1	AZ-2011-9050313	2011-01-03	Summer Hayward	854
2	AZ-2011-6674300	2011-01-04	Devin Huddleston	140
3	BN-2011-2819714	2011-01-04	Mary Parker	27
4	BN-2011-2819714	2011-01-04	Mary Parker	17



Descriptive Statistics

	Recency	Monetary
count	8047	8047.000000
mean	2013-04-19 12:25:40.748104704	291.845657
min	2011-01-01 00:00:00	3.000000
25%	2012-06-08 00:00:00	48.000000
50%	2013-06-11 00:00:00	117.000000
75%	2014-04-30 00:00:00	313.000000
max	2014-12-31 00:00:00	6517.000000
std	NaN	485.212156

	Frequency	Customer Name
count	8047	8047
unique	4117	792
top	AZ-2014-7040665	Joel Peters
freq	11	32



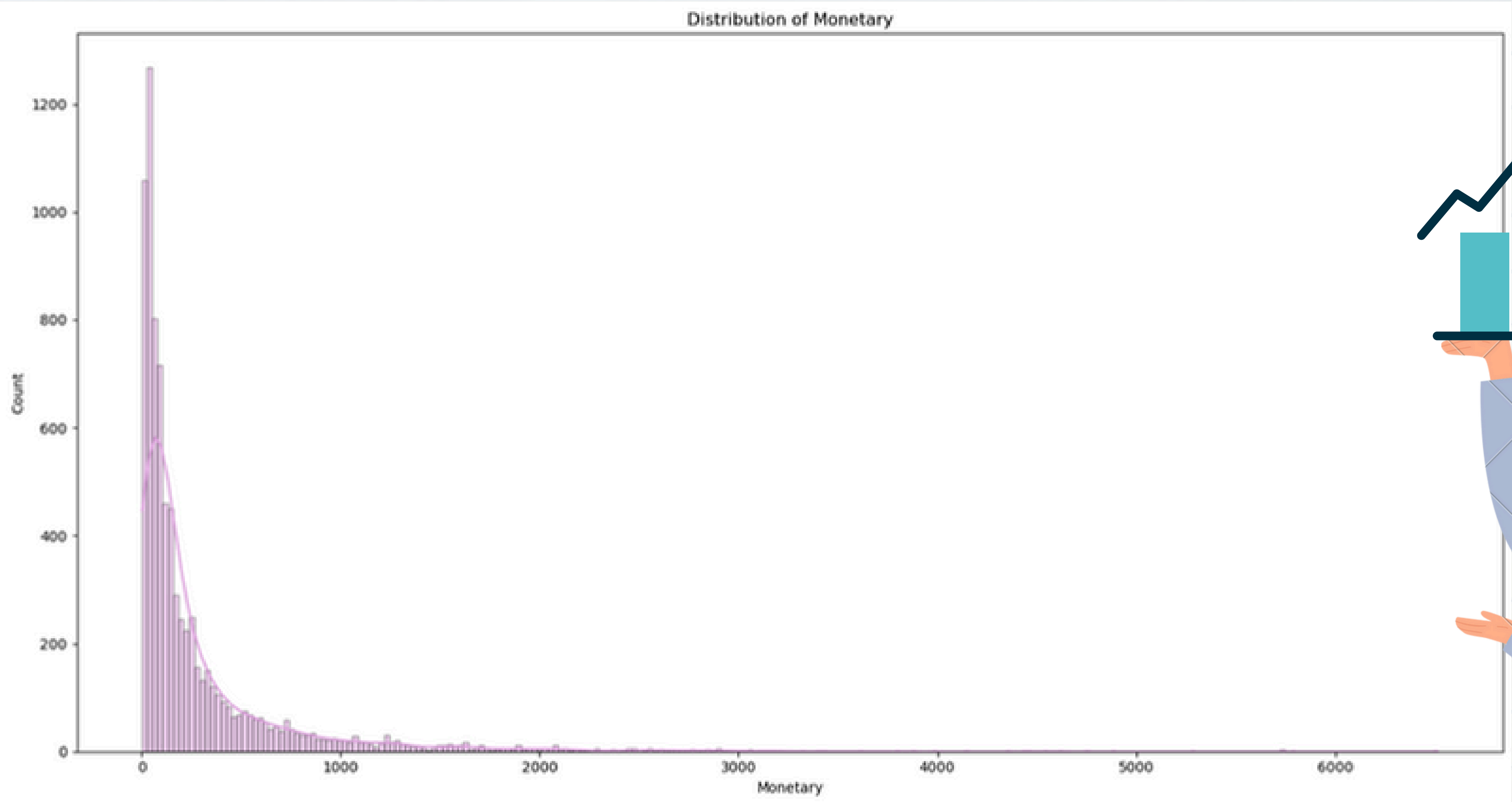


Top 5 Person with Highest Monetary

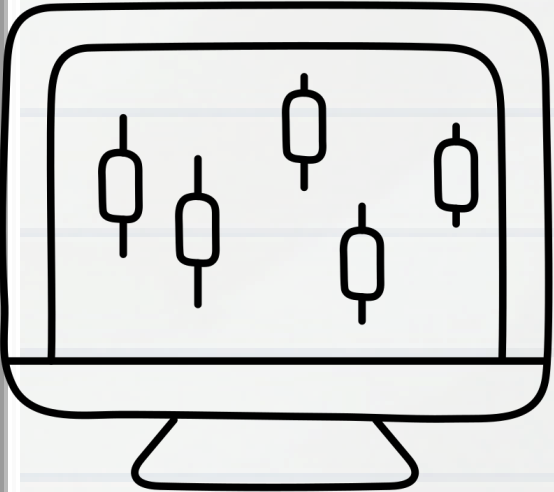
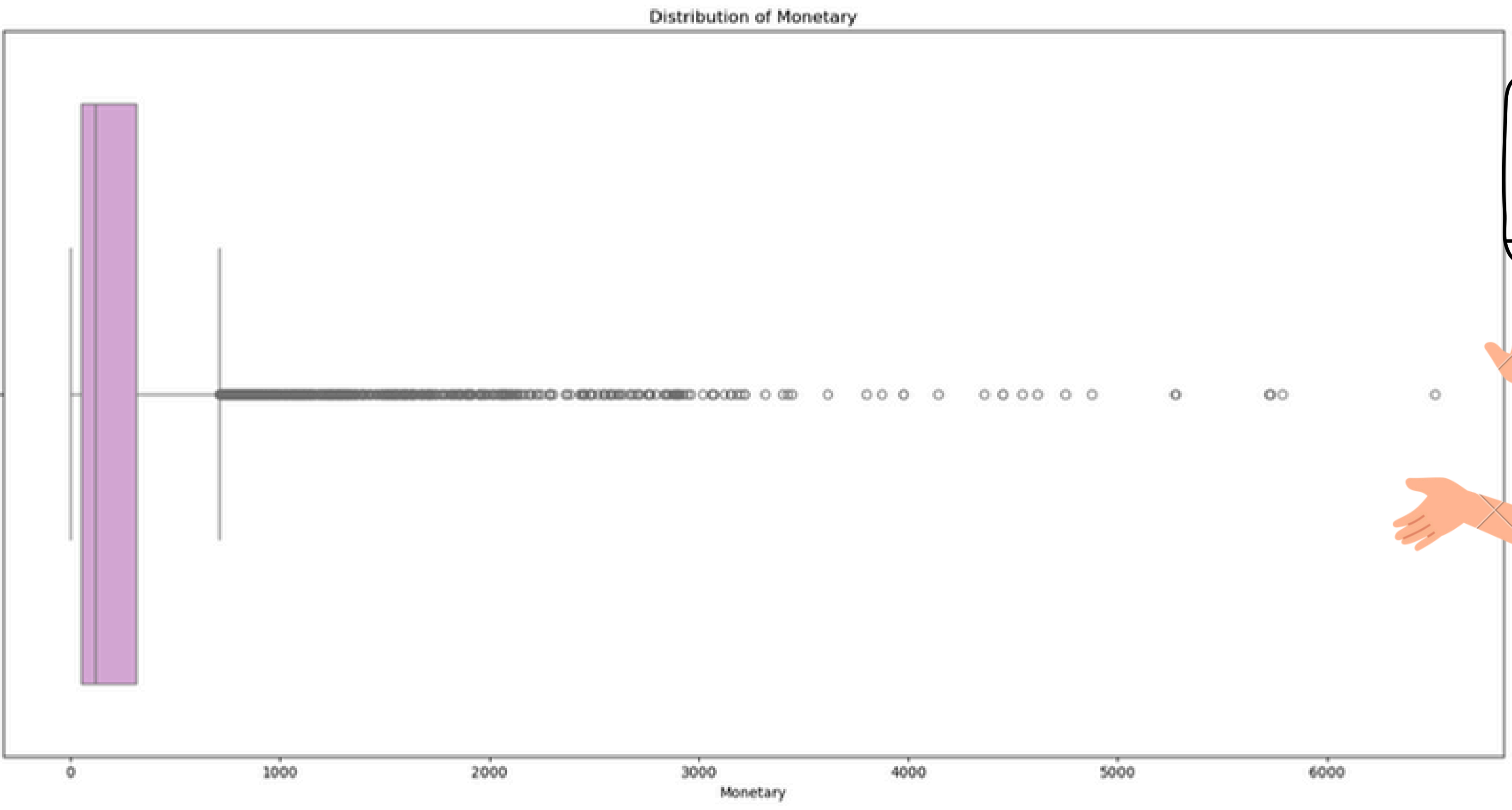
	Frequency	Recency	Customer Name	Monetary
442	AZ-2011-2948812	2011-06-06	Eva Butler	104
986	BN-2011-188092	2011-09-27	Peter Daly	117
1099	BN-2011-4260093	2011-11-01	Casey Forte	87
1273	AZ-2011-6683192	2011-12-03	Gracie Hicks	36
1315	AZ-2011-9732880	2011-12-10	Sofia Goddard	34



Histogram Distribution of Monetary



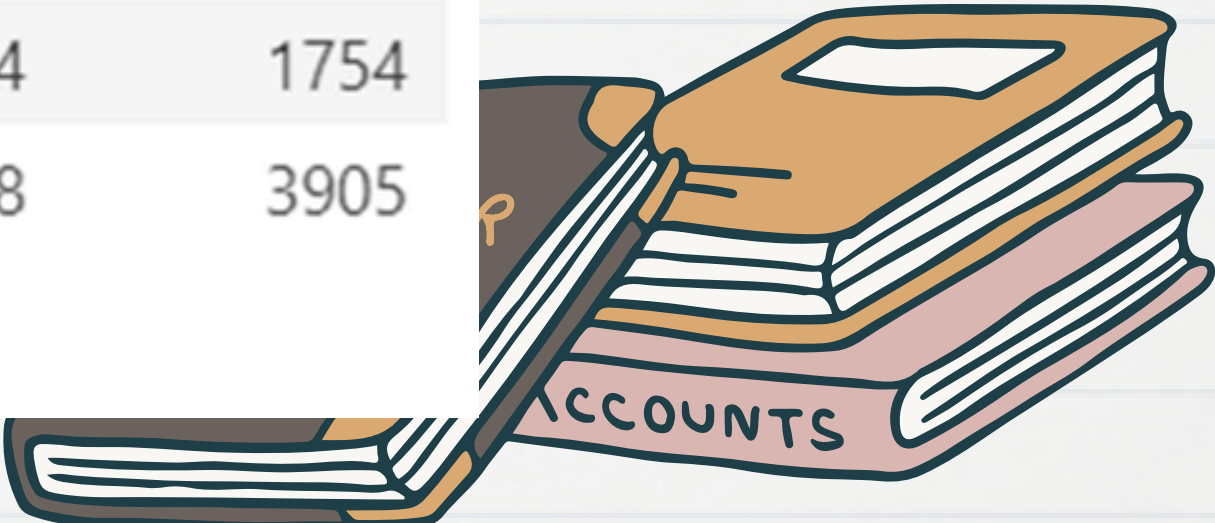
Boxplot Distribution of Monetary



Recency, Frequency, and Monetary (RFM) Table

	Customer Name	Recency	Frequency	Monetary
0	Aaron Bootman	252	11	4281
1	Aaron Cunningham	211	8	2463
2	Aaron Davey	636	4	2238
3	Aaron Macrossan	1199	1	81
4	Abbie Perry	97	4	1313
...	...	...	...	...
787	Zara Loton	116	3	1614
788	Zara Money	12	5	2862
789	Zoe Lanigan	173	4	3249
790	Zoe Lowin	5	4	1754
791	Zona Meyer	13	8	3905

792 rows x 4 columns



RFM Score Assignment

	Customer Name	Recency	Frequency	Monetary	Recency_Score	Frequency_Score	Monetary_Score
0	Aaron Bootman	252	11	4281	1	4	4
1	Aaron Cunningham	211	8	2463	1	4	4
2	Aaron Davey	636	4	2238	1	2	4
3	Aaron Macrossan	1199	1	81	1	1	1
4	Abbie Perry	97	4	1313	2	2	3

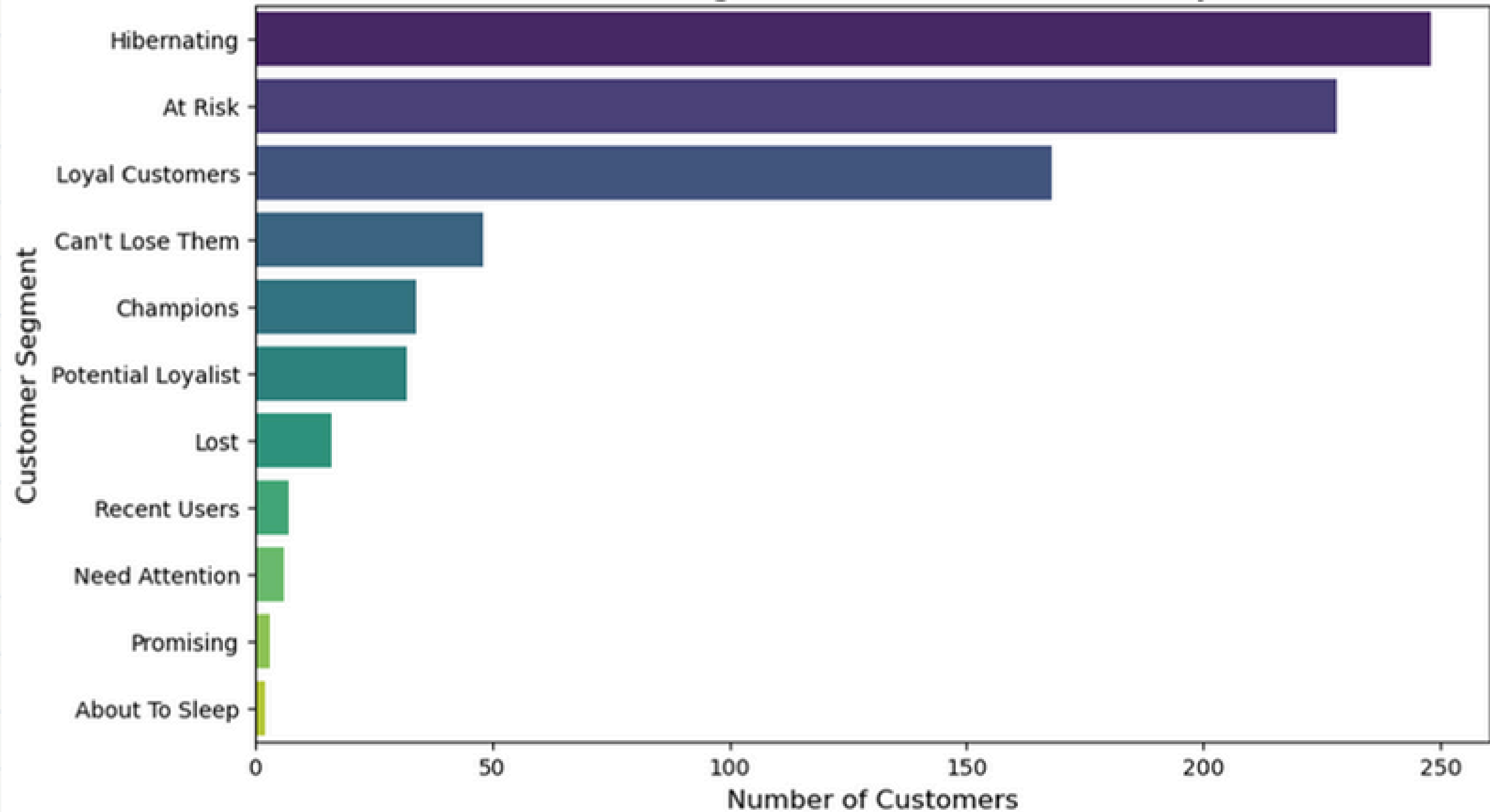


Data Collection for Segmentation

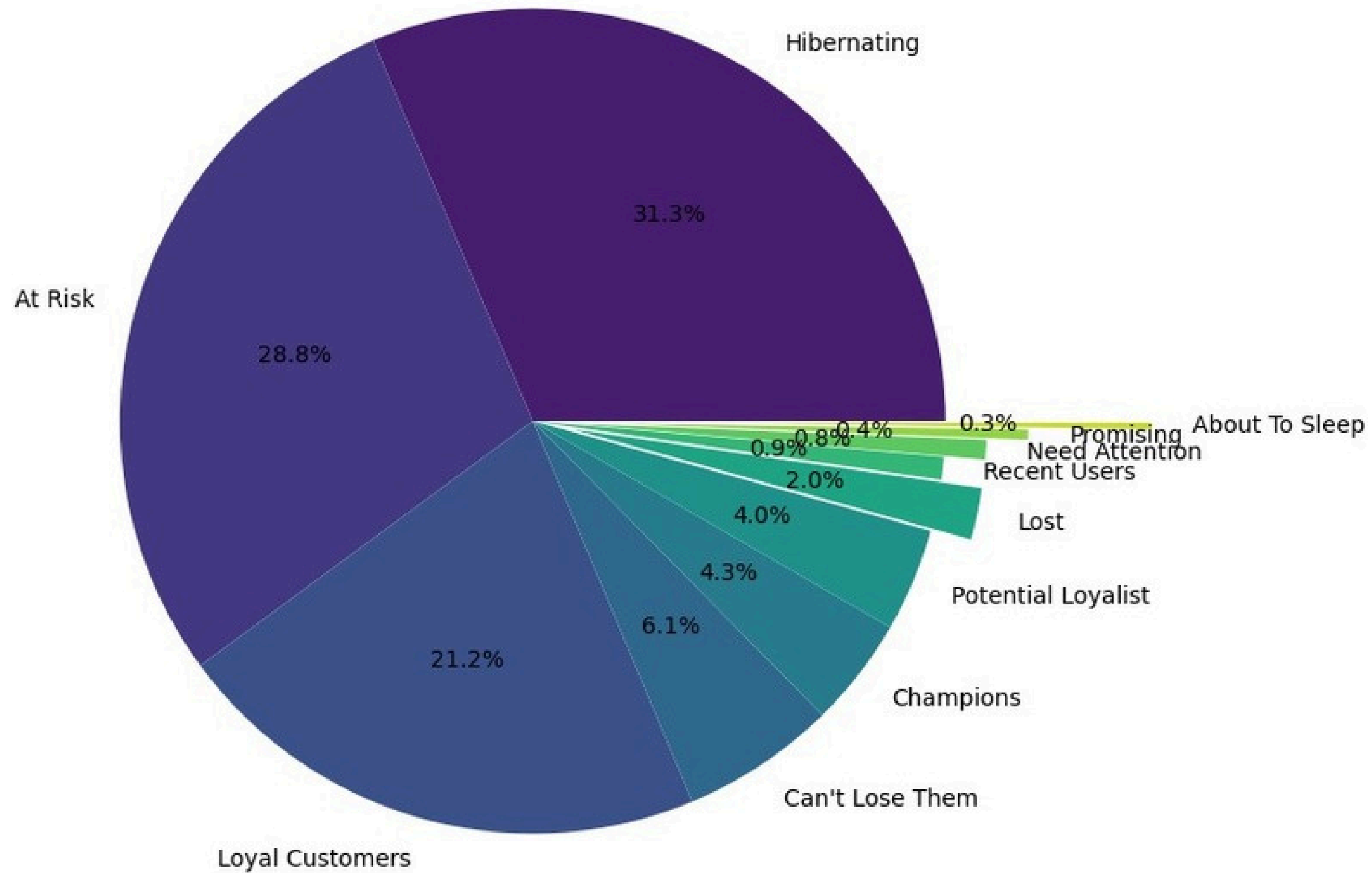
	Customer Name	Recency_Score	Frequency_Score	Monetary_Score	Segment
0	Aaron Bootman	1	4	4	At Risk
1	Aaron Cunningham	1	4	4	At Risk
2	Aaron Davey	1	2	4	Hibernating
3	Aaron Macrossan	1	1	1	Lost
4	Abbie Perry	2	2	3	Hibernating



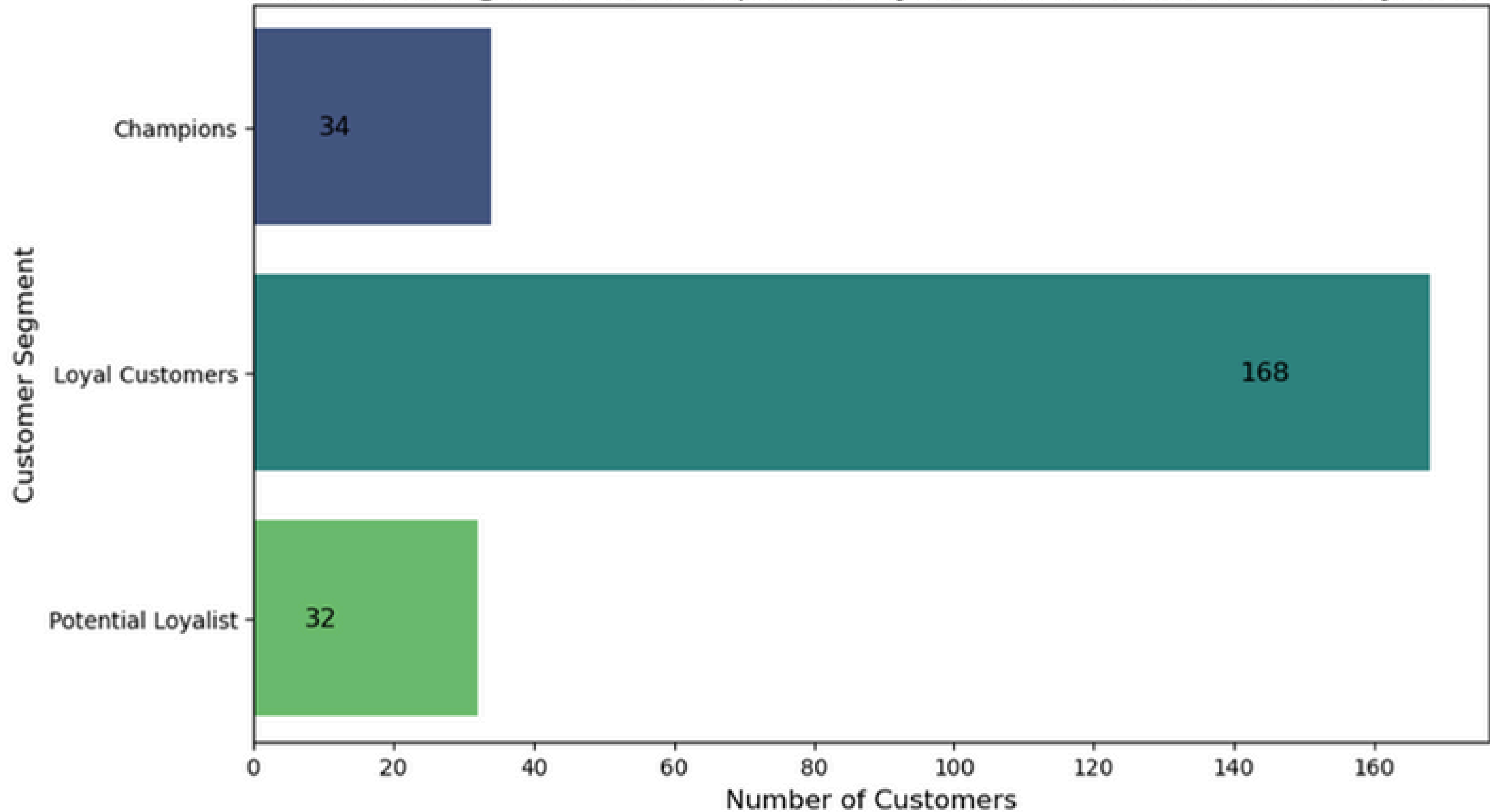
Customer Segments Based on RFM Analysis



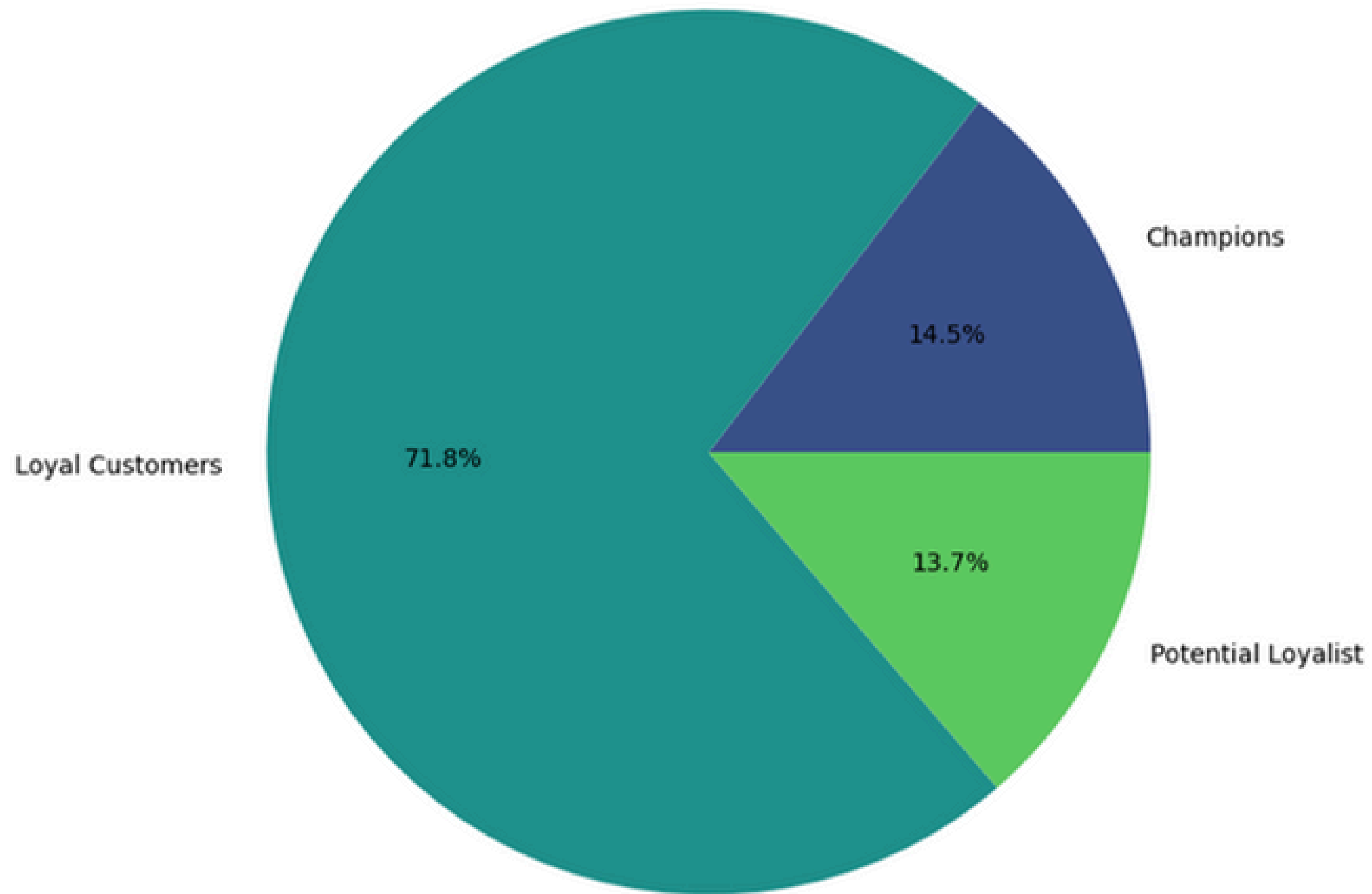
Customer Segments Distribution



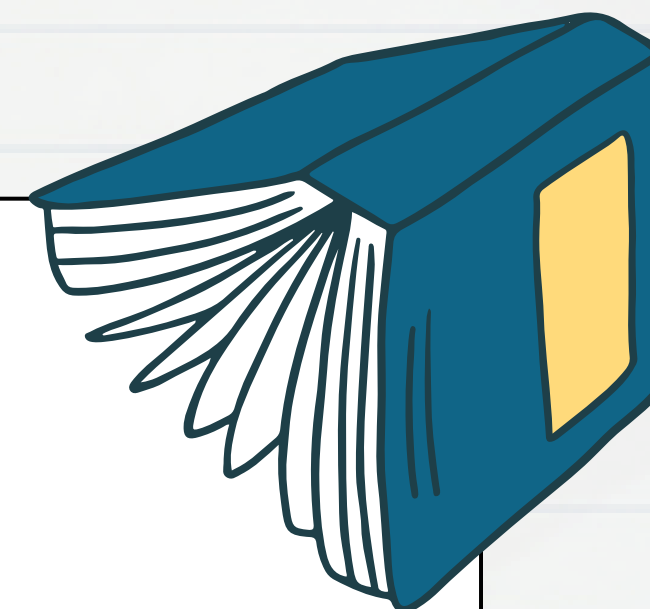
Customer Segments: Champions, Loyal Customers, Potential Loyalists



Customer Segments Distribution: Champions, Loyal Customers, Potential Loyalists



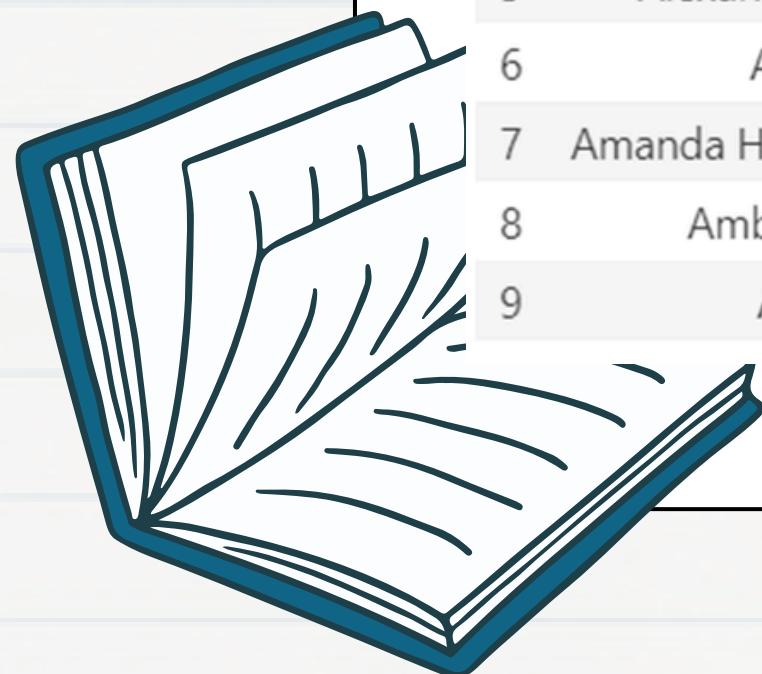
Priority Customer



	Customer Name	Segment
0	Abby Muramats	Loyal Customers
1	Adam Walpole	Loyal Customers
2	Aidan Rowe	Loyal Customers
3	Alex Greenwood	Loyal Customers
4	Alexander Bennett	Loyal Customers
5	Alexandra Portus	Loyal Customers
6	Alta Nokes	Loyal Customers
7	Amanda Hutcherson	Loyal Customers
8	Amber Adams	Loyal Customers
9	Anna Kelly	Loyal Customers

	Customer Name	Segment
0	Adam Harte	Champions
1	Alexis Code	Champions
2	Angelina Hurst	Champions
3	Anna Macdonald	Champions
4	Audrey Knowles	Champions
5	Charles Builder	Champions
6	Craig Rayner	Champions
7	Eloise Sykes	Champions
8	Evie Flockhart	Champions
9	Georgina Garner	Champions

	Customer Name	Segment
0	Alannah Callaway	Potential Loyalist
1	Alice Black	Potential Loyalist
2	Alice Currie	Potential Loyalist
3	Angus Ibsch	Potential Loyalist
4	Archer Hort	Potential Loyalist
5	Ashton Charles	Potential Loyalist
6	Brooke Horton	Potential Loyalist
7	Charlie Power	Potential Loyalist
8	Cynthia Zielinski	Potential Loyalist
9	Erin Gill	Potential Loyalist



Descriptive Statistics for Segmentation

Segment	Recency			Frequency			Monetary		
	mean	count	max	mean	count	max	mean	count	max
About To Sleep	121.500000	2	125	1.000000	2	1	52.000000	2	57
At Risk	229.592105	228	762	6.412281	228	11	3797.083333	228	16047
Can't Lose Them	335.750000	48	1112	3.104167	48	4	783.687500	48	999
Champions	13.970588	34	30	9.088235	34	13	5649.352941	34	13191
Hibernating	273.274194	248	1309	3.487903	248	8	2018.229839	248	8701
Lost	700.812500	16	1412	1.000000	16	1	170.250000	16	470
Loyal Customers	29.696429	168	58	6.279762	168	11	3620.398810	168	10466
Need Attention	77.500000	6	89	3.166667	6	4	712.500000	6	927
Potential Loyalist	73.718750	32	89	6.718750	32	10	4051.687500	32	13056
Promising	48.666667	3	53	2.666667	3	4	595.000000	3	810
Recent Users	10.000000	7	26	2.428571	7	3	512.714286	7	854

Table Summary

Aspek	Champions	Loyal Customers	Potential Loyalists
Recency	Sangat tinggi (baru-baru ini berbelanja)	Tinggi (baru-baru ini berbelanja)	Tinggi (baru-baru ini mulai aktif)
Frequency	Sangat sering	Sering	Sedang (berpotensi meningkat)
Monetary	Nilai transaksi sangat tinggi	Nilai transaksi tinggi	Nilai transaksi sedang
Fokus Strategi	Eksklusivitas, pengalaman premium, menjaga hubungan jangka panjang	Retensi, meningkatkan frekuensi belanja	Meningkatkan frekuensi dan retensi, konversi menjadi pelanggan loyal
Peran dalam Bisnis	Pelanggan terbaik, advocate bisnis	Pelanggan setia, berkontribusi besar	Potensi pelanggan setia di masa depan

Actionable Recommendation



Champions

Pelanggan yang paling sering berbelanja, baru saja melakukan pembelian, dan memiliki nilai transaksi tinggi

Eksklusivitas & Pengalaman Premium



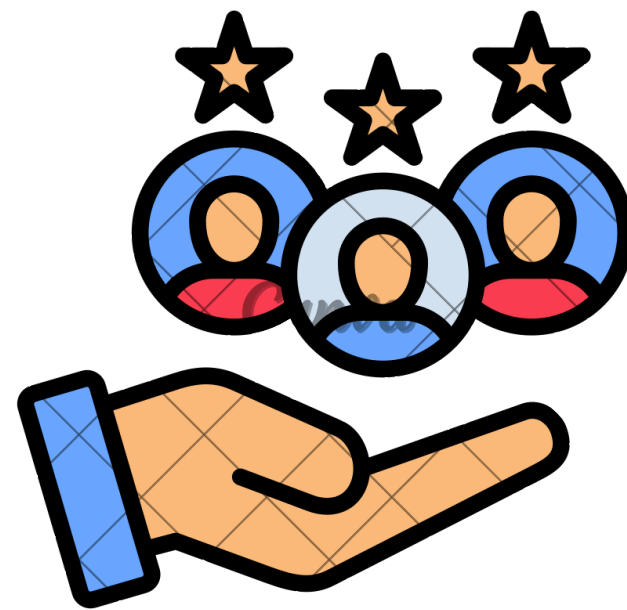
EXTRA
DISCOUNT



Program Loyalitas Premium & Reward besar



Actionable Recommendation



Loyal Customer

Pelanggan yang sering berbelanja, memiliki nilai signifikan terhadap, namun tidak sebesar Champions. Mereka Setia, tetapi tidak berbelanja sebanyak segmen Champions.

Menjaga retensi & Meningkatkan frekuensi belanja



Program loyalitas pengumpulan poin



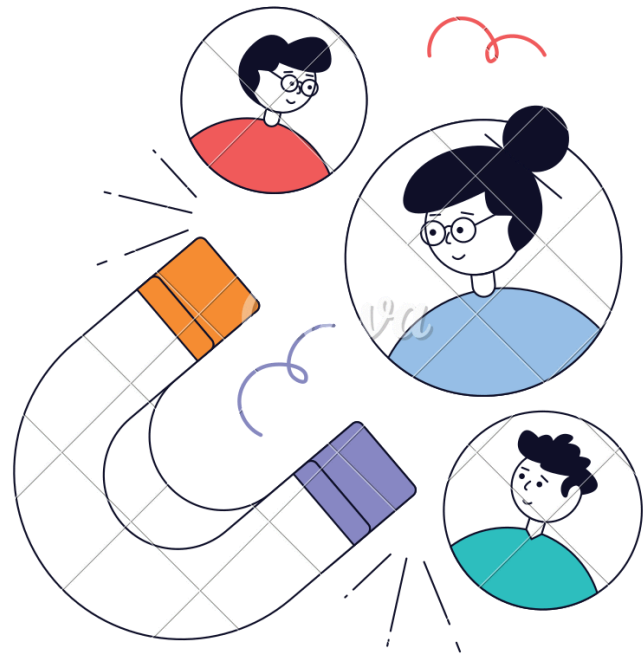
Penawaran berkala khusus pelanggan loyal



Melakukan survey dan mendapatkan feedback

Pelanggan Setia yang berkontribusi besar terhadap bisnis dan perlu dipertahankan agar tetap menjadi pelanggan aktif dan berpotensi naik ke kategori Champions

Actionable Recommendation



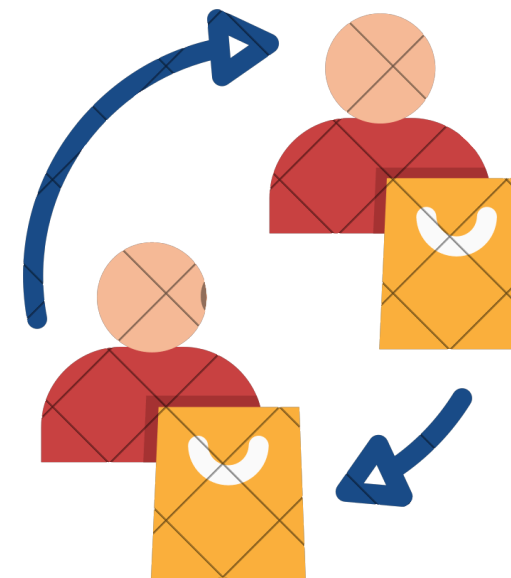
Potential Loyalist

Pelanggan yang baru saja mulai berbelanja secara aktif dan berpotensi menjadi pelanggan loyal di masa mendatang jika diberikan insentif yang tepat.

Meningkatkan retensi & frekuensi belanja



Personalized product Recommendation



Menciptakan program untuk meningkatkan engagement & Retensi



Free shipping or eksklusif promotions

Pelanggan potensial yang menunjukkan tanda-tanda akan menjadi loyal, sehingga memerlukan perhatian khusus untuk memastikan mereka tetap terlibat dan tidak berpindah ke kompetitor

Thank You

